

2-oxoglutarate triggers assembly of active dodecameric *Methanosarcina mazei* glutamine synthetase

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Abstract

Glutamine synthetases (GS) are central enzymes essential for the nitrogen metabolism across all domains of life. Consequently, they have been extensively studied for more than half a century. Based on the ATP dependent ammonium assimilation generating glutamine, GS expression and activity are strictly regulated in all organisms. In the methanogenic archaeon *Methanosarcina mazei*, it has been shown that the metabolite 2-oxoglutarate (2-OG) directly induces the GS activity. Besides, modulation of the activity by interaction with small proteins (GlnK₁ and sP26) has been reported. Here, we show that the strong activation of *M. mazei* GS (GlnA₁) by 2-OG is based on the 2-OG dependent dodecamer assembly of GlnA₁ by using mass photometry (MP) and single particle cryo-electron microscopy (cryo-EM) analysis of purified strep-tagged GlnA₁. The dodecamer assembly from monomers/dimers occurred without any detectable intermediate oligomeric state and was not affected in the presence of GlnK₁. The 2.39 Å cryo-EM structure of the dodecameric complex in the presence of 12.5 mM 2-OG demonstrated that 2-OG is binding between two monomers. Thereby, 2-OG appears to induce the dodecameric assembly in a cooperative way. Furthermore, the active site is primed by an allosteric interaction cascade caused by 2-OG-binding towards an adaption of the transition state catalytic conformation. In the presence of additional glutamine, strong feedback inhibition of GS activity was observed. Since glutamine dependent disassembly of the dodecamer was excluded by MP, feedback inhibition most likely relies on an allosteric binding of glutamine to the catalytic site. Based on our findings, we propose that under nitrogen limitation the induction of *M. mazei* GS into a catalytically active dodecamer is not affected by GlnK₁ and crucially depends on the presence of 2-OG.

eLife assessment

This manuscript reports a **valuable** new mechanism of regulation of the glutamine synthetase in the archaeon *Methanosarcina mazei* and clarifies the direct activation of glutamine synthetase activity by 2-oxoglutarate, thus introducing a novel understanding of how 2-oxoglutarate serves as a central indicator of carbon and nitrogen sensing. The authors provide **solid** evidence using mass photometry, specific activity measurements, and single particle cryo-EM data. This study is of interest to biologists working on the regulation of metabolism.

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Introduction

Nitrogen is one of the key elements in life and it is essentially required in form of ammonium for biomolecules such as proteins or nucleic acids. Two major pathways of ammonium assimilation in bacteria and archaea are known. Under nitrogen (N) sufficiency glutamate dehydrogenase (GDH) is active and generates glutamate from oxoglutarate and ammonium (reviewed in [van Heeswijk et al., 2013](#)). However, under N limitation low ammonium concentrations lead to inactive GDH as a result of its low ammonium affinity, whereas the expression of glutamine synthetase (GS) is strongly induced in response to N limitation ([Bolay et al., 2018](#) ; [Gunka and Commichau, 2012](#) ; [Stadtman, 2001](#)). Consequently, under low ammonium conditions GS together with glutamate synthase (GOGAT) are responsible for ammonium assimilation via the GS/GOGAT pathway, one of the major intersections in central carbon and N metabolism. Accordingly, GS present across all domains of life plays a central role in cellular N assimilation under low N availability. The enzyme, its structure and regulation has been investigated in detail in different organisms for more than half a century (e.g. [Dos Santos Moreira et al., 2019](#) ; [Stadtman, 2001](#) ; [Woolfolk and Stadtman, 1967](#)).

Most of the GS are grouped into three major classes based on their monomeric size and oligomerization properties (overview in [Dos Santos Moreira et al., 2019](#)). GSI and GSIII, both found in bacteria and archaea mostly form dodecamers, whereas GSII found in Eukaryotes form decamers of smaller subunits ([Dos Santos Moreira et al., 2019](#) ; [He et al., 2009](#) ; [Valentine et al., 1968](#) ; [van Rooyen et al., 2011](#)). The GSI class can be further grouped into Ia-type GS and Ib-type GS based on their amino acid sequence and respective molecular mechanisms of activity regulation. Ib-type GS contain a conserved adenylation site (Tyr397 residue near the active site) that allows for covalent modification of Ib-type GS and leads to inactivation of the enzyme ([Brown et al., 1994](#) ; [Magasanik, 1993](#) ; [Shapiro and Stadtman, 1970](#)), whereas Ia-type GS are not covalently modified and mainly show feedback inhibition by end products of the glutamine metabolism including glutamine ([Fisher, 1999](#) ; [Gunka and Commichau, 2012](#)).

GS regulation on transcriptional level

Since in contrast to GDH, GS catalyzed generation of glutamine requires ATP, most organisms strictly regulate the expression of GS in response to the nitrogen availability on the transcriptional level. In gram negative bacteria mainly transcriptional activation of the coding gene (*glnA*) under low nitrogen availability occurs via a transcriptional activator (e.g. NtrC in *Escherichia coli* ([Jiang et al., 1998](#))). For several gram positive bacteria however, the mechanism of regulation is a de-repression of *glnA* transcription under N limitation, which has also been shown for methanoarchaea ([Cohen-Kupiec et al., 1999](#) ; [Fedorova et al., 2013](#) ; [Fisher, 1999](#) ; [Fisher and Wray, 2008](#) ; [Hauf et al., 2016](#) ; [Weidenbach et al., 2010](#) , 2008). Whereas in gram positives

the signal perception is complex and often also involves protein interactions of GS with transcriptional regulators (reviewed in Gunka and Commichau, 2012 [\[1\]](#)), signal perception and transduction in methanoarchaea occurs directly via the small effector molecule 2-oxoglutarate (2-OG), which increases under N limitation. It has been shown that binding of 2-OG to the global N repressor protein NrpR significantly changes the repressor conformation resulting in dissociation from its respective operator (Lie et al., 2007 [\[2\]](#); Weidenbach et al., 2010 [\[3\]](#); Wisedchaisri et al., 2010 [\[4\]](#)). In addition to expression regulation the activity of GS is also strictly regulated in all organisms in response to changing N availabilities, however the underlying molecular mechanism(s) of inhibition significantly differ for the various GS classes and in various organisms (Reitzer, 2003 [\[5\]](#))

Regulation of GS activity: highly diverse and often complex in various organisms

An extensive repertoire of cellular control mechanisms regulating GS activity in response to N availability has been observed in different organisms. Inhibitory mechanisms in response to an upshift range from feedback inhibition by e.g. glutamine or other end products of the glutamine metabolism (e.g. *E. coli* (Stadtman, 2004 [\[6\]](#)), *Bacillus subtilis* (Deuel et al., 1970 [\[7\]](#)), yeast (Legrain et al., 1982 [\[8\]](#))), proteolytic degradation (yeast, (Legrain et al., 1982 [\[8\]](#)), covalent modification by adenylation of the I β -type GS subunits (e.g. enterobacteriaceae), thiol-based GS regulation (e.g. in soybean nodules (Masalkar and Roberts, 2015 [\[9\]](#))), inhibition by regulatory proteins (e.g. in gram positive bacteria (Travis et al., 2022a [\[10\]](#))), inhibition by interactions with small proteins (e.g. inhibitory factors in Cyanobacteria (García-Domínguez et al., 1999 [\[11\]](#); Klähn et al., 2018 [\[12\]](#), 2015 [\[13\]](#))), to directly effecting the activity through the presence or absence of the small metabolite 2-OG, which has been shown for the first time for *Methanosarcina mazei* (Ehlers et al., 2005 [\[14\]](#)). Moreover, often complex regulations for GS activity including several of the different regulatory mechanisms are reported for one organism. For example, yeast GS (ScGS) is regulated via feedback inhibition by glutamine and is susceptible to proteolytic degradation under N starvation. It was also found to assemble into nanotubes (He et al., 2009 [\[15\]](#)) and under advanced cellular starvation into inactive filaments (Petrovska et al., 2014 [\[16\]](#)). In *E. coli*, the activity of the I β -type GS (EcGS) is controlled by cumulative feedback inhibition and covalent modification (reviewed in Reitzer, 2003 [\[5\]](#)). It has been shown that each of the 12 subunits can be modified by adenylation (Tyr397) resulting in an inactivation of the respective subunit (Stadtman, 1990 [\[17\]](#)). Moreover, the adenylation of single subunits in the complex in addition makes the other subunits more susceptible to cumulative feedback inhibition by various substances (Stadtman, 1990 [\[17\]](#)). These substances either bind the glutamine-binding pocket or have an allosteric binding site (Liauw et al., 1993 [\[18\]](#); Woolfolk and Stadtman, 1967 [\[19\]](#)). The dodecameric structure of EcGS has been known for a long time (Almassy et al., 1986 [\[20\]](#); Yamashita et al., 1989 [\[21\]](#)). However, when artificially exposed to divalent cations (Mn²⁺, Co²⁺) it randomly aggregates and produces long hexagonal tubes (paracrystalline aggregates) (Valentine et al., 1968 [\[22\]](#)). The detailed structural information on the mechanisms of this GS-filament formation to an inactive form of EcGS, often associated with stress responses, as well as the reversion into individual active dodecamers has only recently been described by cryo-electron microscopy (cryo-EM) analysis (Huang et al., 2022 [\[23\]](#)). The *B. subtilis* GS has been shown to be feedback regulated. In addition, binding of the transcriptional repressor GlnR to the feedback inhibited complex not only activates the transcription repression function of GlnR (Fisher and Wray, 2008 [\[24\]](#)) but also stabilizes the inactive GS conformation potentially changing from a dodecamer into a tetradecameric structure (Travis et al., 2022a [\[10\]](#)).

In *M. mazei*, a mesophilic methanoarchaeon, which is able to fix N₂, regulation of the central N metabolisms has been studied extensively on the transcriptional and post-transcriptional level (Jäger et al., 2009 [\[25\]](#); Prasse and Schmitz, 2018 [\[26\]](#); Veit et al., 2005 [\[27\]](#)). A central role of 2-OG for the perception of changes in N availabilities has been proposed as has been demonstrated for cyanobacteria (Forchhammer, 1999 [\[28\]](#); Herrero et al., 2001 [\[29\]](#)). The activity of *M. mazei* GS encoded by *glnA₁*, is regulated by several different mechanisms. GS /GlnA₁) is not covalently modified in

response to N availability and thus represents a $\text{I}\beta$ -type-GS (Ehlers et al., 2005 [\[4\]](#)). It has been demonstrated to get directly activated under N starvation due to the high intracellular concentrations of the metabolite 2-OG which directly induces GlnA₁ activity (Ehlers et al., 2005 [\[4\]](#)). 2-OG represents the internal signal for N limitation since under N starvation the internal 2-OG level significantly increases due to missing consumption by GDH (*M. mazei* contains the oxidative TCA part, anabolic). The increased cellular 2-OG concentration has been shown to be directly perceived by the GlnA₁ most likely by direct binding resulting in strong activation (Ehlers et al., 2005 [\[4\]](#)). Besides, we showed first evidence that in addition two small proteins interact with *M. mazei* GlnA₁, the PII-like protein GlnK₁ and small protein sP26 comprising 23 amino acids (Ehlers et al., 2005 [\[4\]](#); Gutt et al., 2021 [\[5\]](#)). The presence and potential interaction of both small proteins show small effects on the GlnA₁ activity, however compared to the strong 2-OG stimulation only to a very low extend, which might be neglectable and due to the indirect GS activity assay. Moreover, based on initial complex formation analysis using size exclusion chromatography (SEC) and pull-down approaches first indications were obtained that in the absence of 2-OG the GlnA₁/GlnK₁ complexes are more stable than in the presence of high 2-OG. This led to the conclusion that due to the shift to N sufficiency after a period of N limitation, GlnA₁ activity is reduced due to the lower 2-OG concentration but also due to the inhibitory protein interaction with GlnK₁ (Ehlers et al., 2005 [\[4\]](#)). Very recently the first structural analysis of *M. mazei* GlnA₁ was reported, showing first GS complexes with GlnK₁ (Schumacher et al. 23). Based on their findings Schumacher et al. propose a regulation of GlnA₁ activity by oligomeric modulation, with GlnK₁ stabilizing the dodecameric structure and the formation of GlnA₁ active sites. Since this proposed model is entirely missing the effects of 2-OG on GlnA₁ activity, we here aimed to study the obtained effects regulating *M. mazei* GlnA₁ activity in more detail by evaluating oligomerization and complex formation between GlnA₁, GlnK₁ and sP26 in dependence of 2-OG employing mass photometry (MP) allowing molecular weight distribution of single complexes in solution and by high resolution cryo-EM.

Results

2-OG is responsible for GlnA₁-dodecamer formation in *M. mazei*

The strep-tagged purified GlnA₁ was analyzed by SEC in the presence of 12.5 mM 2-OG demonstrating that GS is exclusively present in a dodecameric structure, no other oligomers were detectable (suppl. Fig. S1). To investigate the effects of 2-OG on *M. mazei* GlnA₁ in more detail, we employed MP, a method that allows to measure the molecular weight distribution in solution. Strep-tagged purified GlnA₁ (after SEC) was dialyzed into a 2-OG free HEPES buffer (see Materials and Methods) and subsequently analyzed by MP, demonstrating that in the presence of low 2-OG concentrations (0.1 mM) all of the *M. mazei* GlnA₁ was exclusively present as monomers/dimers with no higher molecular weight complexes present. After addition of 12.5 mM 2-OG, the size distribution shifted towards a higher molecular weight complex of 630-700 kDa (calculated based on the measured dimer-size in each measurement; expected molecular weight of dodecamer: 634 kDa) (Fig. 1A, B [\[4\]](#)). This molecular weight corresponds to a fully assembled dodecamer species, the same oligomeric structure that is adapted in GS from other prokaryotes. Using 2-OG concentrations varying between 0.1 and 12.5 mM, complex analysis showed that up to 62 % of all particles were assembled in a dodecamer. This further allowed to determine the binding affinity of GlnA₁ to 2-OG to be $K_D = 0.75 \pm 0.01$ mM 2-OG (based on two biological replicates, calculated with the percentage of dodecamer) as described in Materials and Methods, and verified that no other intermediate oligomeric complexes were detectable during dodecameric assembly (Fig. 1A, C [\[4\]](#), suppl. Fig. S2A, B). Activity measurements of Strep-GlnA₁ in the presence of increasing 2-OG concentrations showed a strong increase of the activity from 0.0 U/mg in the absence of 2-OG up to 7.8 ± 1.7 U/mg in the presence of 12.5 mM 2-OG (six independent protein purifications, Fig. 1D [\[4\]](#)). Thus, we conclude that 2-OG acts as a trigger for dodecameric assembly of *M. mazei* GlnA₁, setting it apart from other bacterial and eukaryotic enzyme variants. Moreover, most likely in addition to

the dodecameric assembly, 2-OG is additionally required for a further 2-OG induced conformational switch of the active site, since saturated GlnA₁ activities are not reached in the presence of 5 mM 2-OG, when most of the GlnA₁ is in a dodecameric structure. For full activity, the presence of 12.5 mM 2-OG is required.

GlnK₁ has no detectable effects on GS dodecamer assembly or activity under the tested conditions

Previous studies have shown protein interactions between *M. mazei* GlnA₁ and GlnK₁ as well as GlnK₁ induced effects on GlnA₁ activity. Consequently, we next tested the effects of GlnK₁ presence on the GlnA₁ oligomerization in the presence of 2-OG. Performing the MP analysis under the tested conditions as before but in the presence of purified GlnK₁ demonstrated that (i) in the absence of 2-OG varying ratios between GlnA₁ and GlnK₁ (20:1, 2:1, 2:10 based on monomers) did not result in any dodecamer assembly of GlnA₁ (Fig. 2A, B), (ii) no difference in the GlnA₁ dodecameric assembly in the presence of 2-OG was obtained in the presence of purified GlnK₁ (2:1), (iii) nor was binding of GlnK₁ to GlnA₁ detected by a respective increase in the mass of the higher oligomeric complex (Fig. 2B, C). Moreover, the presence of GlnK₁ (2:1) neither had an influence on the 2-OG affinity (K_D (-GlnK₁) = 1.06 mM 2-OG; K_D (+ GlnK₁) = 1.02 mM 2-OG, K_D calculated based on the dodecamer/dimer ratio), nor in any ratio on the specific activity of GlnA₁ (Fig. 2D, C: exemplarily showing 2:1; suppl. Fig. S2C, D). Consequently, we conclude that under the conditions tested using purified proteins, GlnA₁ dodecamer assembly occurs independently of GlnK₁ and no binding of GlnK₁ to the dodecameric GlnA₁ occurs. However, we cannot exclude that cellular components/metabolites not present in these experiments are crucial for a GlnA₁-GlnK₁ interaction.

Structural basis of oligomer formation by 2-OG

To now unravel the structural mechanism underlying *M. mazei* GlnA₁ activation by 2-OG, we employed cryo-EM and single-particle analysis. Treating freshly purified Strep-GlnA₁ with 12.5 mM 2-OG, effectively shifted the equilibrium towards fully assembled homo-oligomers as depicted in the MP experiments. In the micrographs, fully assembled ring-shaped particles are visible. However, initial attempts to obtain a 3D reconstruction were hindered by the pronounced preferred orientation of particles within the ice, a challenge which has been overcome by introducing low concentrations of CHAPSO (0.7 mM). In our final dataset, all particles exhibited well-distributed oligomers in diverse orientations. Leveraging this dataset, we aligned the particles to a 2.39 Å resolution structure, revealing well resolved side chains that facilitated seamless model building (Fig. 3, suppl. Fig. S3, suppl. Tab. S2). Consequently, we achieved a structure demonstrating excellent geometry and density fitting.

The detailed structural analysis uncovered that GlnA₁ assembles into a dodecamer characterized by stacked hexamer rings. A single GlnA₁ protomer is composed of 15 β-strands and 15 α-helices and is split into a larger C-domain and an N-domain by helix α3. The dodecameric arrangement is achieved through two distinct interfaces, the hexamer interfaces and inter-hexamer interfaces.

Hexamer interfaces are situated between subunits within each ring, while inter-hexamer interfaces occur between subunits derived from adjacent rings (Fig. 4A, B, C). The structures are highly similar to Gram-positive bacterial GS structures (PDB: 4lnn, Murray et al., 2013), with root mean squared deviations (rmsds) of 0.5–1.0 Å.

A closer inspection of the density reveals the density for the bound 2-OG at an allosteric site localized at the interface between two GlnA₁ protomers in vicinity of the GlnA₁ catalytic site (Fig. 4B, D). Several residues are contributing to its binding. R172' and S189' coordinate the γ-Carboxy-group. Additionally, two tightly bound water molecules are detectable in the binding site.

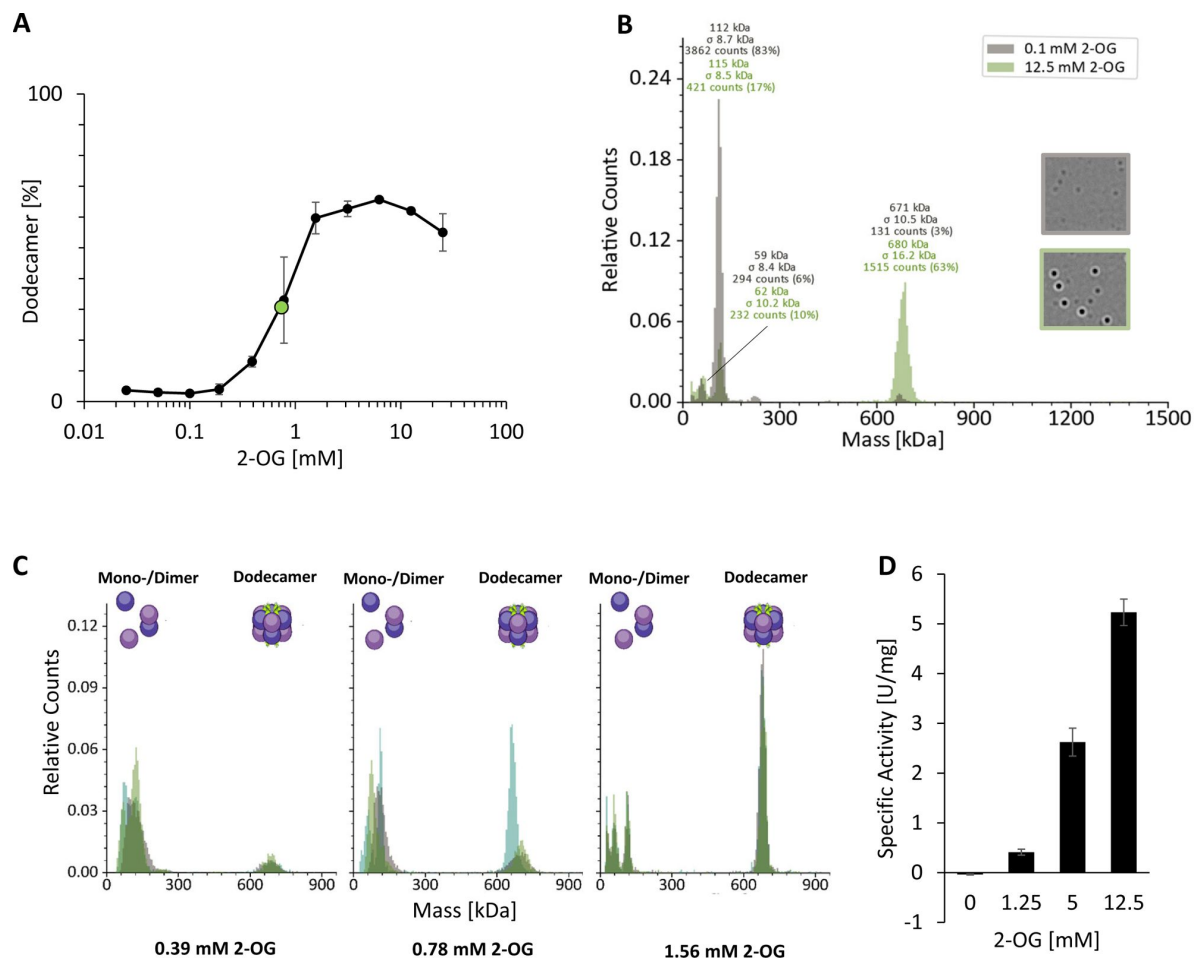


Figure 1

GlnA₁-dodecamer-assembly is induced by 2-OG without detectable oligomeric intermediates.

Oligomerisation states of purified strep-tagged GlnA₁ were assessed in dependence of 2-OG by mass photometry as described in MM using a Refeyn twoMP mass photometer (Refeyn Ltd., Oxford, UK). Mass spectra are shown with relative counts (number of counts in relation to the total counts) plotted against the molecular weight. **A:** 75 nM GlnA₁ were preincubated in the presence of varying 2-OG concentrations (0 to 25 mM) for ten min at room temperature and kept on ice until measurement. The percentage of dodecamer considering the total number of counts was plotted against the 2-OG concentration. One out of two independent biological replicates with each three technical replicates is shown exemplarily. The molecular masses shown above the peaks correspond to a Gaussian fit of the respective peak (Gaussian fit not shown) and the K_D is indicated in green. **B:** Exemplary mass spectra of GlnA₁ oligomers in the presence of 0.1 and 12.5 mM 2-OG. **C:** Mass spectra of the three technical replicates (different green colors) of GlnA₁-oligomers at 0.39, 0.78 and 1.56 mM 2-OG, excluding the presence of intermediates. **D:** The specific activity of purified strep-tagged GlnA₁ was determined as described in MM in the presence of varying 2-OG concentrations (0, 1.25, 5 and 12.5 mM). The standard deviation of four technical replicates is indicated for one out of two biological replicates.

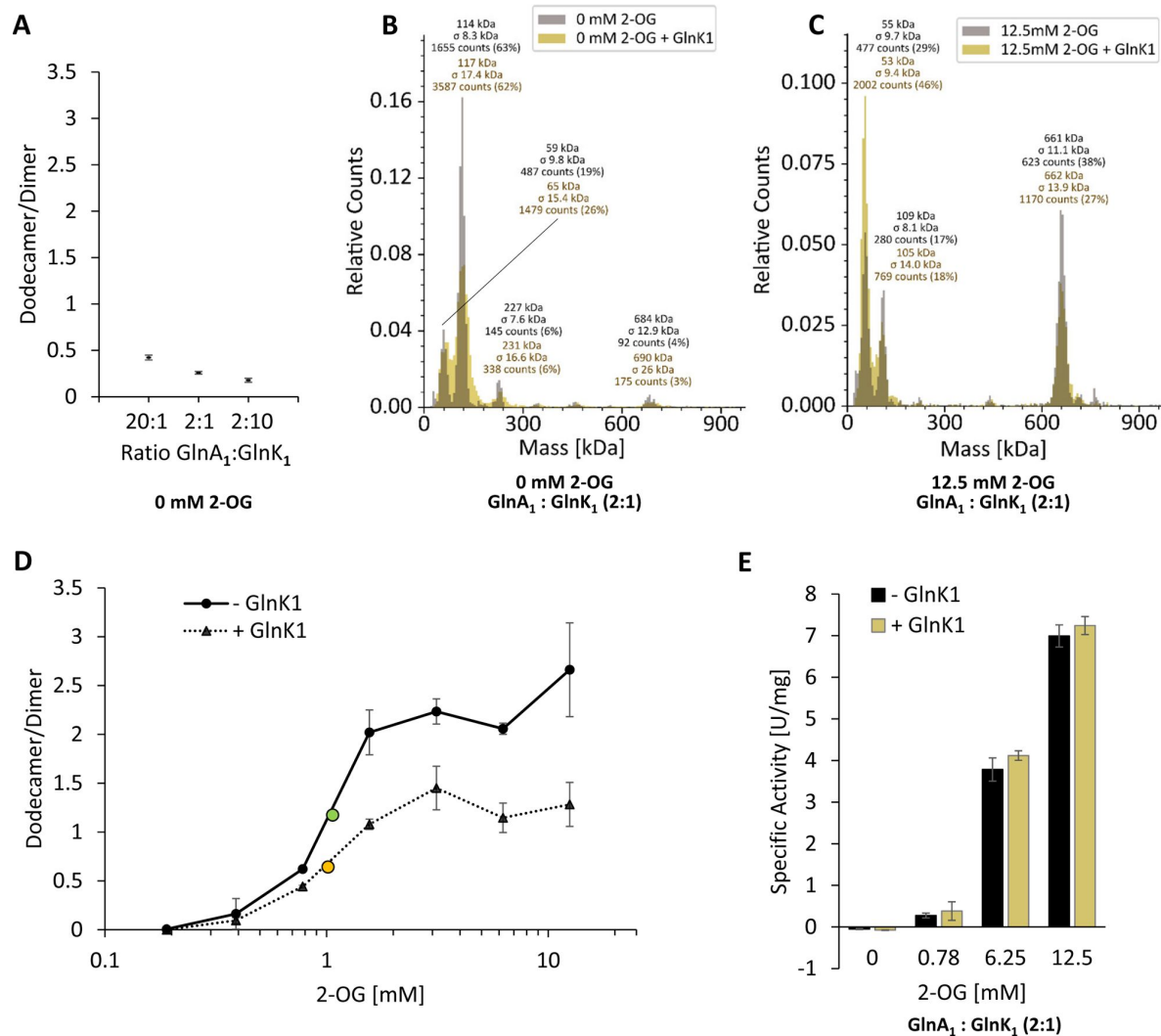


Figure 2

GlnA₁-dodecamer-assembly and activity are not influenced by GlnK₁ under the conditions tested.

Purified strep-tagged GlnA₁ and tag-less GlnK₁ were incubated in the absence or presence of 2-OG in varying concentrations for ten min at RT. Oligomerisation states were assessed by mass photometry. Mass spectra are shown with relative counts (see Fig. 1). **A**: The obtained ratio of GlnA₁ dodecamer/dimer of three technical replicates are shown for varying ratios between GlnA₁ and GlnK₁ (20:1, 2:1, 2:10, ratios relating to monomers) in the absence of 2-OG. **B**, **C**: Exemplary mass spectra of GlnA₁ incubated in the absence and presence of GlnK₁ (2:1) at 2-OG concentrations of 0 mM (**B**) and 12.5 mM (**C**). The molecular masses shown above the peaks correspond to a Gaussian fit of the respective peak (Gaussian fit not shown). **D**: 200 nM monomeric GlnA₁ were preincubated with GlnK₁ (in a 2:1 ratio) in the presence of varying 2-OG concentrations (0.19 to 12.5 mM) for ten min at RT. The percentage of GlnA₁ dodecamer considering the total number of counts was plotted against the 2-OG concentration. One biological replicate with three technical replicates was performed. The ratio of GlnA₁ dodecamer/dimer was plotted against the 2-OG concentration and the KD is indicated in green (●, -GlnK₁) and yellow (●, + GlnK₁). **E**: The specific activity of purified strep-tagged GlnA₁ in the absence and presence of GlnK₁ (ratio 2:1) was determined as described in MM in the presence of varying 2-OG concentrations (0, 0.78, 6.25 and 12.5 mM). The standard deviations of four technical replicates of one biological replicate are indicated.

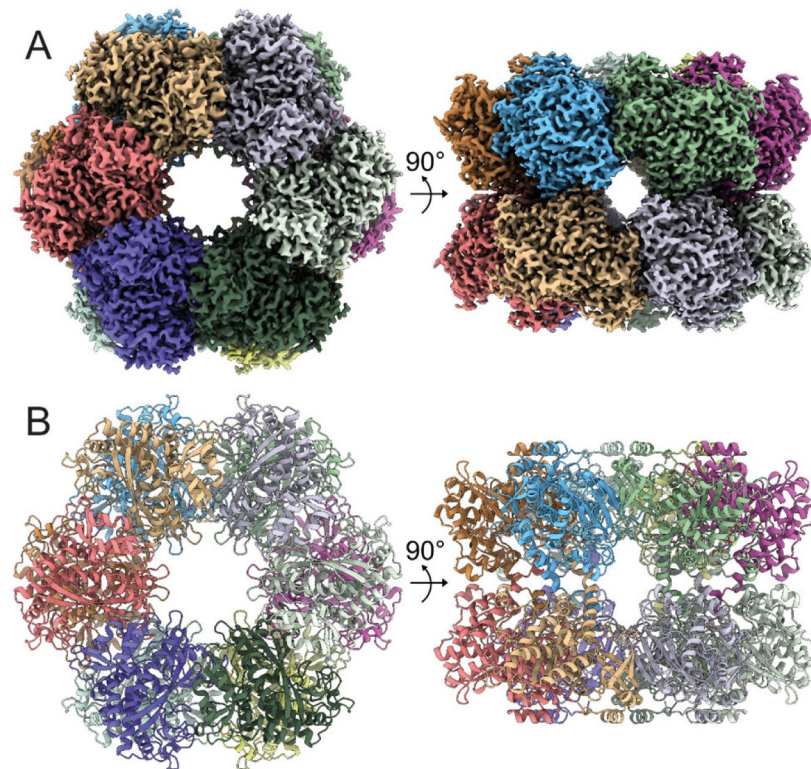


Figure 3

Structure of *M. Mazei* GlnA₁ with 2-OG.

A: Three-dimensional segmented cryo-EM density of the dodecameric complex colored by subunits. **B:** Corresponding views of the GlnA₁ atomic model in cartoon representation.

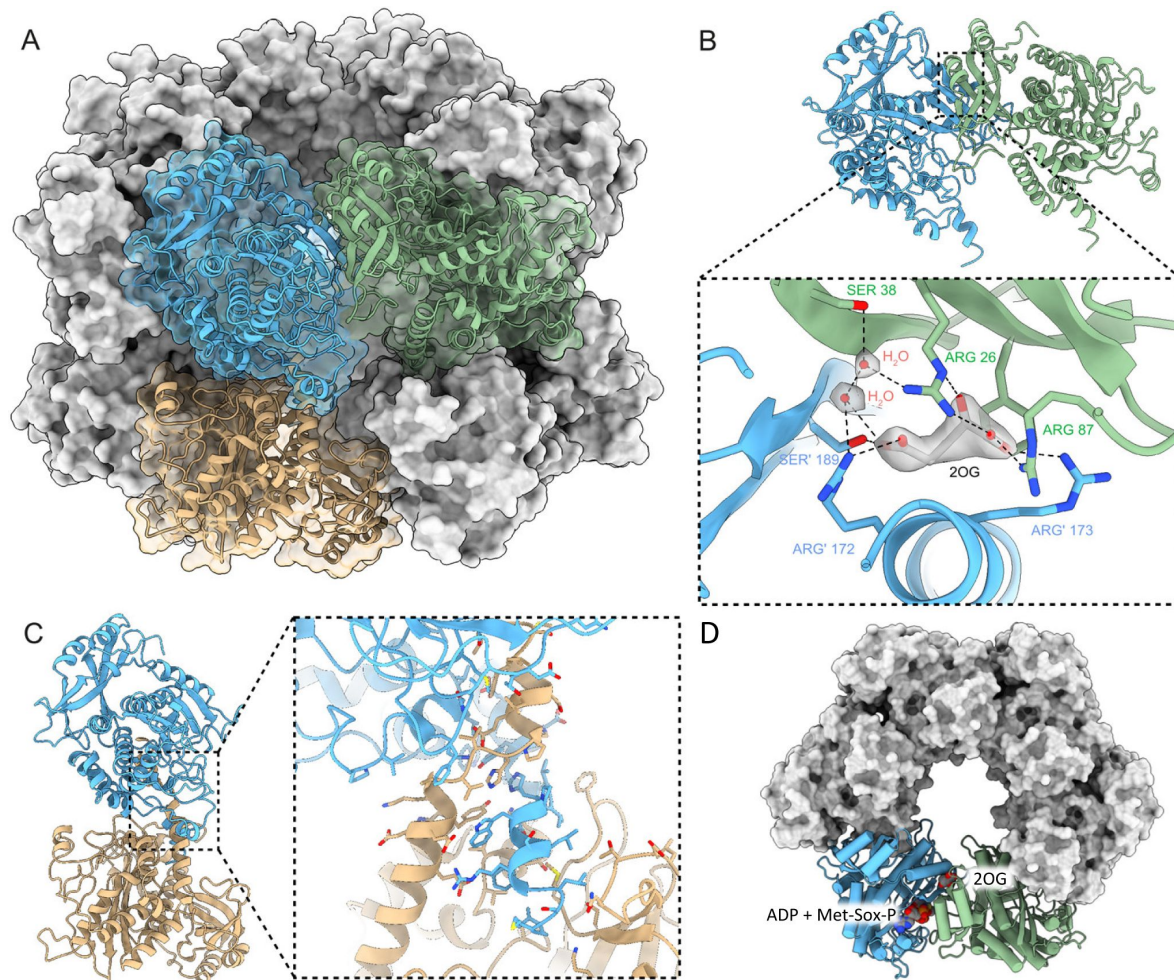


Figure 4

Dimeric Interface and 2-OG binding site of dodecameric GlnA₁.

A: Surface representation of the *M. mazei* GlnA₁ 2-OG dodecamer with three GlnA₁ protomers fitted in cartoon representation into the dodecamer as vertical (blue and ochre) and horizontal (blue and green) dimers. **B:** Horizontal dimers and close-up of 2-OG binding site. Important residues are shown as atomic stick representation, primed labels indicate neighboring protomer. 2-OG and water molecules important for ligand binding fitted into density are shown in grey. Dotted lines represent polar interactions between 2-OG, waters and residues. **C:** Vertical dimers and close-up of dimerization site. C-terminal helices H14/15 and H14'/ H15' of two neighboring protomers lead to tight interaction, mediated by hydrophobic and polar interactions. **D:** Top-view of GlnA₁ hexamer, 2-OG and substrate binding sites are depicted for one horizontal dimer.

One is interacting with the γ -Carboxy group while being stabilized by another water that is coordinated by S38 and R26. Latter arginine is coordinating the α -Keto-group and, together with R87 and R173', the α -Carboxy group of 2-OG (**Fig. 4B**). Notably, F24 stabilizes the 2-OG via stacking with its phenyl ring. This binding contribution from two GlnA₁ protomers at the intersubunit junction enhances activation by boosting readiness and the rate of full complex assembly. It operates akin to molecular glue that facilitate the observed cooperative assembly.

A comparison with the substrate-bound GlnA₁ structure (PDB: 8tfk, [Schumacher et al. 2023](#)) revealed that the catalytically important residues in *M. mazei* are the aspartic acid (D57) that abstracts the proton from ammonium and the catalytic glutamic acid, Glu307. The active site of *M. mazei* GlnA₁ is formed at the interface between two subunits in the hexamer and formed by five key catalytic elements surrounding the active site: the E flap (residues 303–310), the Y loop (residues 369–377), the N loop (residues 235–247), the Y* loop (residues 152–161) and the D50' loop (residues 56–71). The latter one is the only one that originates from adjacent neighboring protomer (**Fig 5C, E**).

Superposition of our structure with the apo-*M. mazei* X-ray structure (PDB: 8tfb, [Schumacher et al., 2023](#)) reveals that 2-OG binding also triggers further movements that lead to structural changes in the substrate binding pocket (**Fig. 5A, B, D**). R87' and its loop undergo a dramatic flip to coordinate 2-OG and D170 of helix α 3 (residues 167–181) (**Fig. 5A**). This, combined with the action of other coordinating residues, initiates a motion that is propagated through the entire protein. Notably, helix α 3 shifts forward, causing F184 to flip over and facilitate a T-shaped aromatic interaction with F202. The resulting pull on F202 causes F204 to flip, allowing π -stacking with the purine moiety of ATP (**Fig. 5B**). This series of structural changes primes the active site for ATP binding by already adopting the side chain conformations that are observed in analogue (Met-Sox-P-ADP)-bound structure (transition state) (PDB: 8tfk, [Schumacher et al., 2023](#)), thus facilitating nucleotide binding (**Fig. 5C, E**).

Additionally, the D50' loop adopts a position similar to the transition state in a catalytic competent conformation. This involved a remodeling of the loop, leading to the positioning of key catalytic residues in a catalytic competent configuration. Compared to the apo structure ([Schumacher et al., 2023](#)), R66 flips out of the catalytic pocket, now accommodating R319 which participates in phosphoryl transfer catalysis (**Fig. 5D**). In addition, Asp 57' moves deeper into the binding site, facilitating the proton abstraction of NH^+ and preparing for its attack on the phosphorylated glutamate. Similar to the ATP/ADP binding site, these catalytic elements are primed to ideally stabilize the tetrahedral transition state. This is illustrated by the superposition of the inhibitor-bound, transition-state locked structure ([Schumacher et al., 2023](#)) (**Fig. 5C, E**).

Feedback inhibition by glutamine does not affect the dodecameric *M. mazei* GlnA₁ structure

For bacteria it is known, that GS can be feedback inhibited. Very recently, the first feedback inhibition of an archaeal GS by glutamine has been reported for *Methermicoccus shengliensis* GS ([Müller et al., 2023](#)). The specific arginine residue identified to be relevant for the feedback inhibition is R66. Consequently, we generated the respective *M. mazei* GlnA₁ mutant protein changing the conserved arginine to alanine (R66A) (see also **Fig. 5D, E**) and compared the purified strep-tagged mutant protein with the wildtype (wt) protein. In the presence of 12.5 mM 2-OG, the mutant protein showed the same specific activity as obtained for the wt. However, when supplementing 5 mM glutamine, exclusively the wt was strongly feedback inhibited, whereas the R66A mutant protein was not significantly affected (**Fig. 6A**). In *B. subtilis*, R62 is responsible for feedback inhibition. The superposition of the apo-BsGS structure (PDB: 4lnn, [Murray et al., 2013](#)) with our 2-OG-bound GlnA₁ reveals a similar positioning of the respective *M. mazei* R66 (**Fig. 6B**) indicating a similar mechanism. Moreover, we can rule out an effect on the oligomeric

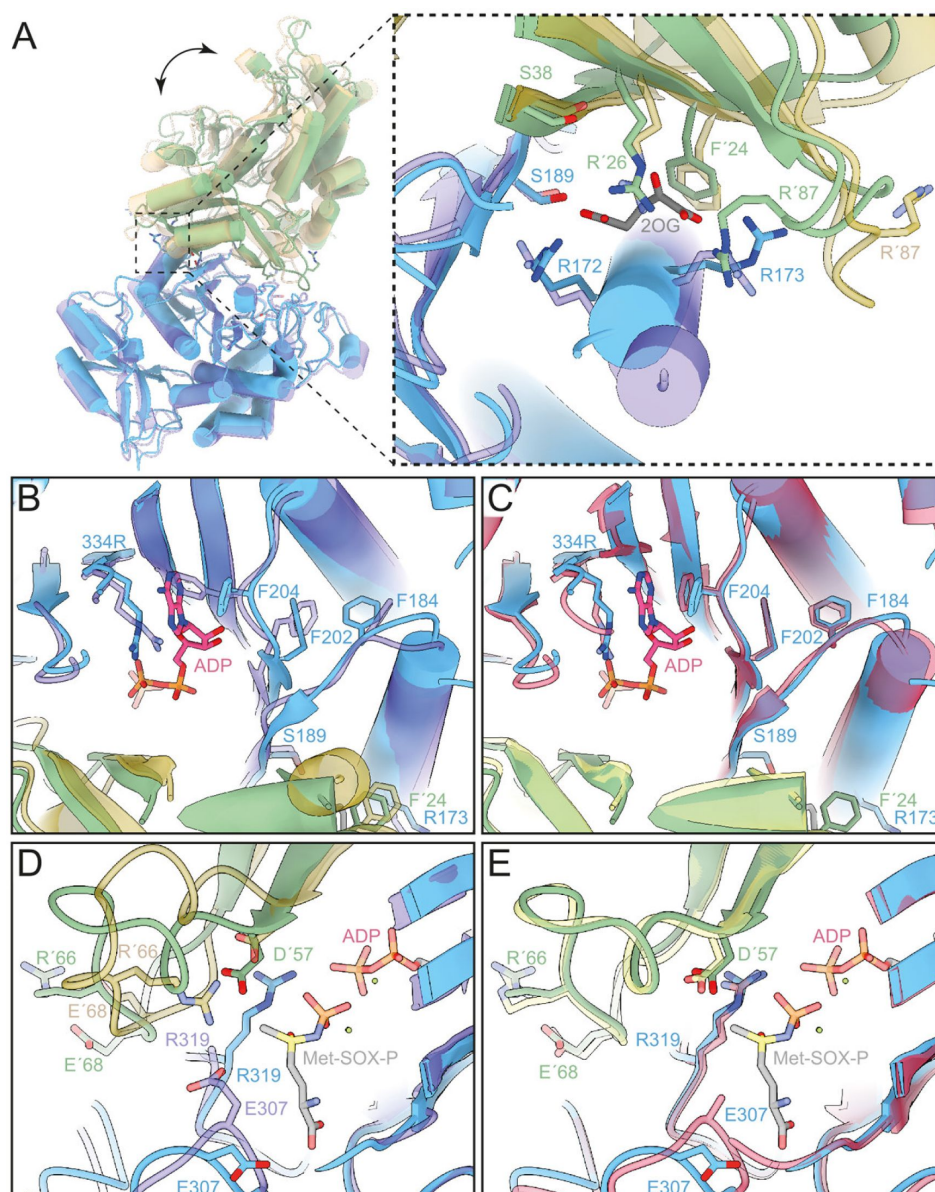


Figure 5

Comparison of 2-OG and substrate binding site of 2-OG bound, apo and TS structures (Schumacher et al., 2023).

Atomic models in cartoon, important residues shown in stick representation. Colors: blue/green, purple/ochre and red/yellow represent *M. mazei* GlnA₁ 2-OG, *M. mazei* GlnA₁ apo (PDB: 8tfb, Schumacher et al., 2023) and *M. mazei* GlnA₁ Met-Sox-P-ADP (PDB: 8tfk, Schumacher et al., 2023) transition state (GlnA₁ TS), respectively. **A left:** GlnA₁ 2-OG dimer in superposition with GlnA₁ apo showing large scale movements upon 2-OG binding. **A right:** Close-up of 2-OG binding site of GlnA₁ 2-OG in superposition with GlnA₁ apo. Dramatic movement of Helix $\alpha 3$ (residue 167-181) and R87 loop show effect of 2-OG binding. **B:** Close-up of substrate binding site of GlnA₁ 2-OG in superposition with GlnA₁ apo and ADP ligand from GlnA₁ TS. Helix $\alpha 3$ movement upon 2-OG binding leads to a cascade of conformational changes of the phenylalanines F184, F202 and F204 that lead to a priming of the active site for ATP binding. **C:** Close-up of substrate binding site of GlnA₁ 2-OG in superposition with GlnA₁ TS shows high similarity between 2-OG bound and transition state structure. **D:** Close-up of substrate binding site of GlnA₁ 2-OG in superposition with GlnA₁ apo and Met-Sox-P ligand from GlnA₁ TS. Large structural changes of the D50-loop with ejection of the R66 key-residue shown. Flipping of the loop allows R319 and D57 to move in further and catalyze phosphoryl-transfer and attack of NH_4^+ , respectively. **E:** Close-up of the substrate binding site of GlnA₁ 2-OG in in superposition with GlnA₁ TS reveals strong similarity between 2-OG bound and transition state structure in the active site.

structure of GlnA₁ by MP analysis, clearly showing that glutamine does not induce disassembly of the dodecameric wt GlnA₁ (Fig. 6C). Instead, this effect can be explained with the role of R66 being an important residue to bind to glutamine in the product state of the enzyme.

Discussion

2-OG is crucially required for *M. mazei* GS assembly to an active dodecamer and induces the conformational state towards an active open state

In *M. mazei* increased 2-OG concentrations act as central N starvation signal (Ehlers et al., 2005). Here we demonstrated the importance of 2-OG as the major regulator of *M. mazei* GlnA₁ activity by using independent methods, MP and cryo-EM, to detect and structurally characterize single complexes with high resolution and quantify the different oligomeric complexes. We have found mono- and dimeric GlnA₁ (apo GlnA₁) to be inactive and crucially require 2-OG to form an active dodecameric complex. Moreover, this dodecameric conformation is the only active state of GlnA₁. In the first step, 2-OG assembles the dodecamer by binding at the interface of two subunits (Fig. 4B) and functions as a molecular glue between neighbouring subunits. The assembly upon 2-OG addition observed using MP appears to be cooperative, fast and without any detectable intermediate states (Fig. 1B, C). Only immediately after thawing a frozen purified GlnA₁ preparation and in case that no additional SEC was performed prior to MP analysis, samples showed additional octameric complexes in MP with low abundance (suppl. Fig. S4). However, octameric complexes were never observed in cryo-EM or detected by SEC analysis of frozen purified GlnA₁ samples. Consequently, octamers are most likely broken or disassembled GlnA₁-dodecamers or dead-ends in assembly with no physiological function, rather than an incomplete dodecamer during assembly. Thus, our findings are contrary to the assembly model proposed by Schumacher et al. (Schumacher et al., 2023).

As a second step of activation, the allosteric binding of 2-OG causes a series of conformational changes in GlnA₁ protomers, which prime the active site for the transition state and hence catalysis of the enzyme. This conformational change of the ATP-binding pocket of the dodecameric GlnA₁ upon 2-OG binding goes hand in hand with the observed increased activity at higher 2-OG concentrations (Fig. 1). Comparing our 2-OG-bound GlnA₁ dodecameric structure and the dodecameric *M. mazei* GlnA₁ transition state (PDB: 8tfk) and apo structures (PDB: 8ftb) reported by Schumacher et al. (Schumacher et al., 2023), clearly demonstrates that 2-OG transfers GlnA₁ into its open transition state conformation (Fig. 5). The conformation of our 2-OG-bound dodecamer resembled the transition state conformation (ADP-Met-Sox-bound complex) reported by Schumacher et al., even though in our case no ATP was added (Fig. 5E). A reconfiguration of the active site upon 2-OG-binding has also been reported for GS in *Methanothermococcus thermolithotrophicus* (Müller et al., 2024). In this report, which does not delineate dodecamer assembly at all, it was demonstrated that binding of 2-OG in one protomer-protomer interface of a dodecameric GS causes a cooperative domino effect in the hexameric ring of *M. thermolithotrophicus* GS (Müller et al., 2024). A 2-OG bound protomer undergoes a conformational change and thereby induces the same shift in its neighbouring protomer (Müller et al., 2024). This is comparable to our observed cooperativity of *M. mazei* dodecamer assembly at low 2-OG concentrations ($K_D = 0.75$ mM, percentage of dodecamer). On the other hand, *M. mazei* GlnA₁ reaches maximal activity only at much higher 2-OG concentrations and likely requires a fully 2-OG-occupied dodecamer for maximal activity. The here obtained high activities by 2-OG saturation (up to 9 U/mg) in comparison with previously described *M. mazei* GlnA₁ activities in the absence of 2-OG in the significantly lower range (mU/mg) (Gutt et al., 2021; Schumacher et al., 2023) support our conclusion that 2-OG is substantial for the GlnA₁ active state.

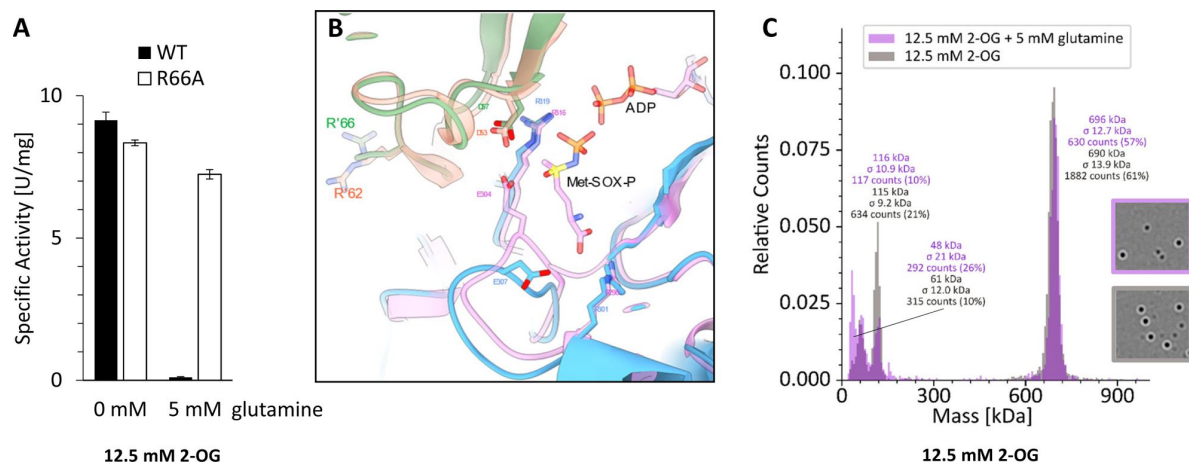


Figure 6

Feedback inhibition of GlnA₁ by glutamine.

A: Specific activity of purified strep-tagged GlnA₁ (wt) and the respective R66A-mutant protein was determined as described in Materials and Methods in the presence of 12.5 mM 2-OG and after additional supplementation of 5 mM glutamine. For wt and the R66A-mutant one out of two biological independent replicates are exemplarily shown, the deviation indicates the average of four technical replicates. **B:** Superposition GS structures without glutamine of *M. mazei* (blue, green) and *B. subtilis* (orange, pink; PDB: 4lnn, Murray et al., 2013): substrate binding-site including R'66 (R'62, respectively), which are responsible for feedback inhibition. **C:** Exemplary mass spectra of Strep-GlnA₁ with 12.5 mM 2-OG in presence and absence of 5 mM glutamine. The molecular masses shown above the peaks correspond to a Gaussian fit of the respective peak (Gaussian fit not shown).

GlnA₁ activity is further regulated by feedback inhibition, small proteins and possibly filament formation

M. mazei GlnA₁ belongs to the group of Ia-type GS, which are known to be feedback inhibited. We confirmed a strong feedback inhibition by a genetic approach and found R66 to be the key residue for this inhibition (Fig. 6) as suggested in Müller et al. 2024. The mechanism of feedback inhibition has been described in detail for *B. subtilis* GS (Murray et al., 2013). There, R62 plays the central role by binding glutamine and inducing a well ordered inactive structure at the substrate-binding pocket upon glutamine-binding (Murray et al., 2013). The homologous *M. mazei* R66 likely conveys a similar way of inhibition to *B. subtilis* GS (Fig. 6B, alignment in suppl. Fig. S5).

Further regulations by the two small proteins sP26 and the PII-like protein GlnK₁ have previously been reported for *M. mazei* (Ehlers et al., 2005; Gutt et al., 2021; Schumacher et al., 2023). However, in the present study neither an interaction with GlnK₁, nor GlnK₁ effects on GlnA₁ complex formation analysed by MP, nor an effect of GlnK₁ on GlnA₁ activity was detectable under the conditions used at varying 2-OG concentrations (0.1 to 12.5 mM) and ratios of GlnK₁ to GlnA₁ (20:1, 2:1, 2:10) (Fig. 2). Moreover, the addition of GlnK₁ did not result in a change of the K_D for 2-OG for the dodecamer GlnA₁ assembly (Fig. 2D). In previous reports, GlnK₁ was shown to interact with GlnA₁ in vivo after a nitrogen upshift by pull-down approaches (Ehlers et al., 2005), pointing towards an inhibitory function of GlnK₁ under shifting conditions from N limitation to N sufficiency. Similarly, we could not determine a cryo-EM structure including sP26 despite adding large excess of the small protein either obtained by co-expression or by addition of a synthetic peptide. Because these attempts were unsuccessful, we speculate that yet unknown cellular factor(s) might be required for an interaction of GlnA₁ with both small proteins, GlnK₁ and sP26, which however is difficult to simulate under in vitro conditions with purified proteins. Taken this into account, we speculate about a potential function of the two small proteins beyond GlnA₁ inactivation or activation. Since the GlnA₁ reaction is coupled to the GOGAT reaction (GS/GOGAT system) and the products of the two reactions replenish the substrates for one other, it is tempting to speculate that GlnA₁ and GOGAT experience metabolic coupling by sP26 and/or GlnK₁ e.g. by being involved in recruiting or separating GOGAT from GlnA₁.

Finally, higher oligomeric states of GS enzymes have been known for a long time for organisms like yeast and *E. coli* (He et al., 2009; Huang et al., 2022; Petrovska et al., 2014; Valentine et al., 1968). Interestingly, dependent on the ice thickness and on higher concentrated areas of the grids, we could also observe filament-like structures of *M. mazei* GlnA₁ in cryo-EM and resolved their structure at a resolution of 6.9 Å (suppl. Fig. S6). Such GlnA₁ filaments are also detectable in the cryo-EM images of Schumacher et al. 2023 but were not reported. Their interface is much alike the previously reported *E. coli* GS filament structures (Huang et al., 2022). The physiological relevance of filamentation in *M. mazei* however remains unresolved and raises the question, whether an additional rapid modulation of GlnA₁ activity through higher oligomeric states exists, as described e.g. for yeast GS most depending on stress conditions (Petrovska et al., 2014).

M. mazei GlnA₁ shows unique properties

Overall, we have confirmed 2-OG to be the central activator of GlnA₁ in *M. mazei*, which assembles the active dodecamer and induces a conformational switch towards an active open state. Though 2-OG has previously been reported as an on-switch for (methano)archaeal GS activity (Ehlers et al., 2005; Müller et al., 2024; Pedro-Roig et al., 2013), the 2-OG-triggered assembly is novel and described exclusively for *M. mazei* GlnA₁. In this respect, the way of GS regulation in *M. mazei* is unique across all prokaryotic GS studied so far. Neither in cyanobacteria, enterobacteria or *Bacillus* is 2-OG a direct activator, nor is complex (dis-)assembly a mode of regulating GS activity in

any other of these model organisms. This is further supported by the absence of up to three of those four arginines - coordinating 2-OG in *M. mazei* GlnA₁ - in these organisms (suppl. Fig. S5). The cyanobacterial, enterobacterial and gram positive GS are present in the cell as active dodecamers (Almassy et al., 1986 [DOI](#); Bolay et al., 2018 [DOI](#); Deuel et al., 1970 [DOI](#)). However, these dodecamers are inactivated upon sudden N sufficiency through very different mechanisms: *Synechocystis* GS is blocked by small proteins (inhibitory factors), the enterobacterial GS experiences gradual adenylation of subunits which abolishes the enzyme activity and *B. subtilis* GS is feedback inhibited by glutamine and further inhibited by binding of the transcription factor GlnR (Almassy et al., 1986 [DOI](#); Bolay et al., 2018 [DOI](#); Klähn et al., 2018 [DOI](#), 2015 [DOI](#); Stadtman, 2001 [DOI](#); Travis et al., 2022b [DOI](#)) (see Fig. 7 [DOI](#)).

The direct 2-OG activation and glutamine feedback inhibition of *M. mazei* GS are two fast, reversible and very direct ways of reacting towards the changing N status of the cell. We propose that the direct activation through 2-OG without any required additional protein as it is the case for all other regulations, is a more simple and direct regulation of GS. Due to the evolutionary placement of methanoarchaea and haloarchaea, where a direct 2-OG regulation has been found exclusively, this may represent an ancient regulation.

Materials and methods

Strains and plasmids

For heterologous expression and purification of Strep-tagged GlnA₁ (MM_0964), the plasmid pRS1841 was constructed. The glnA₁-sequence along with the sP26-sequence (including start-codon: ATG) were codon-optimized for *Escherichia coli* expression and commercially synthesized by Eurofins Genomics on the same plasmid (pRS1728) (Ebersberg, Germany). Polymerase chain reaction (PCR) was performed using pRS1728 as template and the primers (Eurofins Genomics, Ebersberg, Germany) GlnAopt_NdeI_for (5'TTTCATATGGTTCAGATGAAAAATG3') and GlnA1opt_BamHI_rev (5'TTTGGATCCTTACAGCATGCTCAGATAACGG3'). The resulting GlnA₁-opt PCR-product and vector pRS375 were restricted with NdeI and BamHI (NEB, Schwalbach, Germany); the resulting pRS375 vector fragment and the GlnA₁ fragment were ligated resulting in pRS1841. For heterologous expression of Strep-GlnA₁, pRS1841 was transformed in *E. coli* BL21 (DE3) cells (Thermo Fisher Scientific, Waltham, Massachusetts) following the method of Inoue (Inoue et al., 1990 [DOI](#)). For generating the Arg66Ala-mutant, a site-directed mutagenesis was performed. pRS1841 was PCR-amplified using primers sdm_GlnA_R66A_for (5'ATTGAAGAAAGCGATATGAAACTGGCGC3') and sdm_GlnA_R66A_rev (5'CGCGGTAAAGCCCTGAATGCTGCTACC3') by Phusion High-Fidelity polymerase (Thermo Fisher Scientific, Waltham, Massachusetts) followed by religation resulting in plasmid pRS1951. For heterologous expression, pRS1951 was transformed into *E. coli* BL21 (DE3).

In order to co-express sP26 along with Strep-GlnA₁, the construct pRS1863 was generated. pRS1728 with the codon-optimized sP26-sequence and pET21a (Novagen, Darmstadt, Germany) were restricted with NdeI and NotI and the resulting untagged sP26_opt was ligated into the pET21a backbone yielding pRS1863. pRS1863 was co-transformed with pRS1841 into *E. coli* BL21 (DE3) cells (Thermo Fisher Scientific, Waltham, Massachusetts) selecting for both Kanamycin (pRS1841 derived) and Ampicillin (pRS1863 derived) resistance.

The plasmid pRS1672 was constructed for producing untagged GlnK₁. The GlnK₁ gene was PCR-amplified using primers GlnK1_MM0732.for (5'ATGGTTGGCTATGAAATACGTAATTG3') and GlnK1_MM0732.rev (5'TCAAATTGCCTCAGGTCCG3') and cloned into pETSUMO by using the Champion™ pET SUMO Expression System (Thermo Fisher Scientific, Waltham, Massachusetts) according to the manufacturer's protocol. pRS1672 was then transformed into *E. coli* DH5α and BL21 (DE3) pRIL (suppl. Tab. S1).

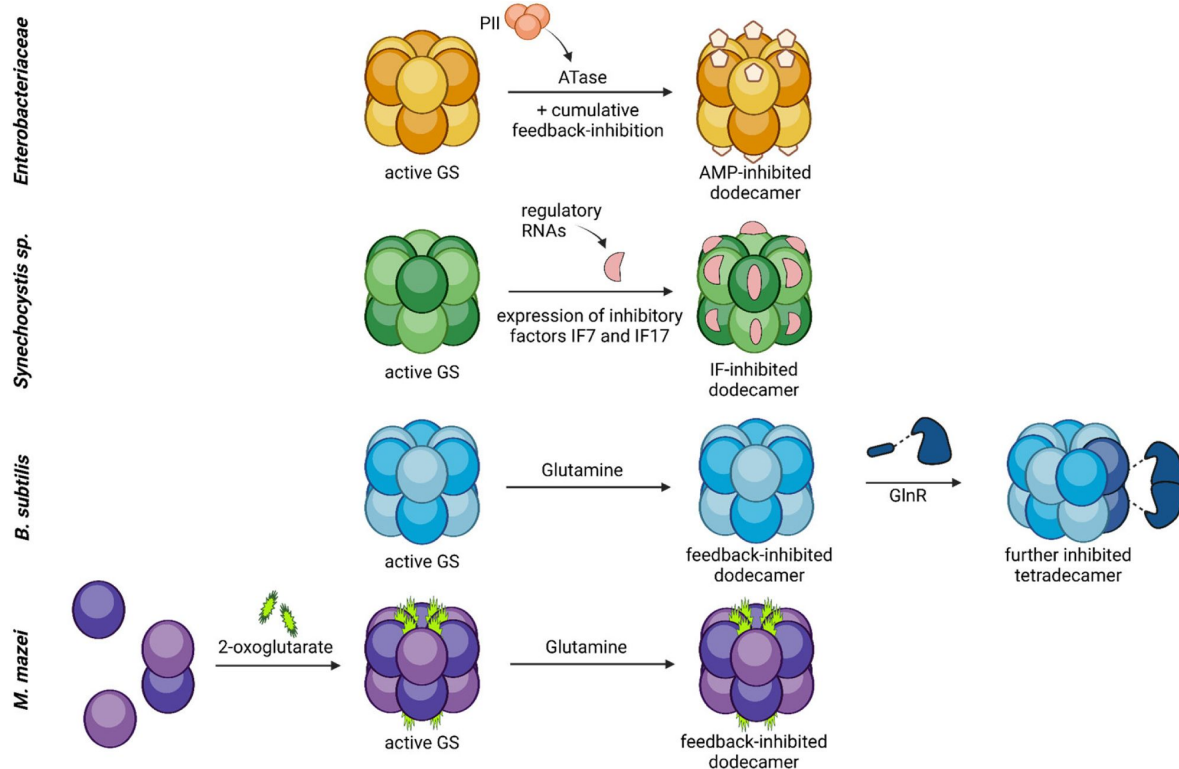


Figure 7

Model of the various molecular mechanisms of glutamine synthetase activity regulation.

Comparison of the regulation of glutamine synthetase activity in *E. coli* /*Salmonella typhimurium*, and *B. subtilis*, *Synechocystis* and *M. mazei*. GS are in general active in a dodecameric, unmodified complex under nitrogen limitation. Upon an ammonium upshift, GS are inactivated by feedback inhibition (BcGS, *E. coli*), covalent modification (adenylylation, EcGS) or binding of (small) inactivating proteins (*Synechocystis*, BsGS). *M. mazei* GS on the contrary is regulated via the assembly of the active dodecamer upon 2-OG-binding and furthermore is strongly feedback inhibited by glutamine. (Bolay et al., 2018 [DOI](#); Klähn et al., 2018 [DOI](#), 2015 [DOI](#); Stadtman, 2001 [DOI](#); Travis et al., 2022b [DOI](#)). Created with [BioRender.com](#)
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Heterologous expression and protein purification: Strep-GlnA₁ and GlnK₁

Heterologous expression of Strep-GlnA₁-variants (pRS1841 and pRS1951) and Strep-GlnA₁-sP26-coexpression (pRS1841 + pRS1863) were performed in 1 l Luria Bertani medium (LB, Carl Roth GmbH + Co. KG, Karlsruhe, Germany). *E. coli* BL21 (DE3) containing pRS1841, pRS1841 and pRS1863 or pRS1951 was grown to an optical turbidity at 600 nm (T_{600}) of 0.6 - 0.8, induced with 25 μ M isopropyl β -D-1-thiogalactopyranoside (IPTG, Carl Roth GmbH + Co. KG, Karlsruhe, Germany) and further incubated over night at 18 °C and 120 rpm. The cells were harvested (6,371 x g, 20 min, 4 °C) and resuspended in 6 ml W-buffer (100 mM TRIS/HCl, 150 mM NaCl, 2.5 mM EDTA, (chemicals from Carl Roth GmbH + Co. KG, Karlsruhe, Germany), 12.5 mM 2-oxoglutarate (2-OG, Sigma-Aldrich, St. Louis, Missouri), pH 8.0). After the addition of DNase I (Sigma-Aldrich, St. Louis, Missouri), cell disruption was performed twice using a French Pressure Cell at 4.135×10^6 N/m² (Sim-Aminco Spectronic Instruments, Dallas, Texas) followed by centrifugation of the cell lysate for (30 min (13,804 x g, 4 °C). The supernatant was incubated with 1 ml equilibrated (W-buffer) Strep-Tactin sepharose matrix (IBA, Gottingen, Germany) at 4°C for 1 h at 20 rpm. Strep-tagged GlnA₁ was eluted from the gravity flow column by adding E-buffer (W-buffer + 2.5 mM desthiobiotine (IBA, Gottingen, Germany)). Strep-GlnA₁ was always purified and stored in the presence of 12.5 mM 2-OG, either in E-buffer or 50 mM HEPES, pH 7.0 at 4 °C for a few days or with 5 % glycerol at -80 °C (chemicals from Carl Roth GmbH + Co. KG, Karlsruhe, Germany).

His₆-SUMO-GlnK₁ was expressed similarly using *E. coli* BL21 (DE3) pRIL + pRS1672. Expression was induced with 100 μ M IPTG, incubated at 37 °C, 180 rpm for 2 h and harvested. The pellet was resuspended in phosphate buffer (50 mM phosphate, 300 mM NaCl, pH 8 (chemicals from Carl Roth GmbH + Co. KG, Karlsruhe, Germany)) and the cell extract was prepared as described above. His-tag-affinity chromatography-purification was performed with a Ni-NTA agarose (Qiagen, Hilden, Germany) gravity flow column, the protein was purified by stepwise-elution with 100 and 250 mM imidazole (SERVA, Heidelberg, Deutschland) in phosphate buffer. SUMO-protease (Thermo Fisher Scientific, Waltham, Massachusetts) was used according to the manufacturer's protocol to cleave the His₆-SUMO-GlnK₁ and obtain untagged GlnK₁ by passing through the Ni-NTA-column after the cleavage. Elution fractions of protein purifications were analyzed on 12 % SDS-PAGE gels and the protein concentrations were determined by Bradford (Bio-Rad Laboratories, Hercules, California) or Qubit protein assay (Thermo Fisher Scientific, Waltham, Massachusetts).

Determination of glutamine synthetase activity

The glutamine synthetase activity was determined by performing a coupled optical assay (Shapiro and Stadtman, 1970 [\[4\]](#)). The assay was performed as described in Gutt et al. 2021 [\[5\]](#) with modifications. Modifications included the use of 50 mM HEPES, the adjustment of ATP-pH to 7.0 and the use of 5 mM glutamine (Sigma-Aldrich, St. Louis, Missouri) in some assays. The assays were performed with four technical replicates per condition including two concentrations of GlnA₁ (2 x 10 μ g and 2 x 20 μ g of Strep-GlnA₁). Strep-GlnA₁ was stored in E-buffer or 50 mM HEPES containing 12.5 mM 2-OG which was dialysed against 50 mM HEPES pH 7 using Amicon® Ultra cartridges with 30 kDa filters (MilliporeSigma, Burlington, Massachusetts) for the enzyme assays in the absence of 2-OG.

Mass photometry

The molecular weight of protein complexes was analysed by mass photometry (MP) using a Refeyn twoMP mass photometer with the AcquireMP software (Refeyn Ltd., Oxford, UK). All measurements were performed in 50 mM HEPES, 150 mM NaCl pH 7.0 (MP-buffer, chemicals from Carl Roth GmbH + Co. KG, Karlsruhe, Germany) on 1.5 H, 24 x 60 mm microscope coverslips with Culture Well Reusable Gaskets (GRACE BIO-LABS, Bend, Oregon). Strep-GlnA₁ and untagged GlnK₁ were prepared as described above. Prior to MP experiments, a size exclusion chromatography

(SEC) was performed with GlnA₁ in the presence of 12.5 mM 2-OG on a Superose™ 6 Increase 10/300 GL column (Cytiva, Marlborough, Massachusetts) with a flow rate of 0.5 ml/min. Only the dodecameric fraction was used for MP experiments and dialysed against MP buffer using Amicon® Ultra catridges with 30 kDa filters (MilliporeSigma, Burlington, Massachusetts) beforehand. The Gel Filtration HMW Calibration Kit (Cytiva, Marlborough, Massachusetts) was used as a standard in SEC. 75 – 200 nM monomeric Strep-GlnA₁ were used in the MP measurements, GlnK₁ was added accordingly in the desired ratio calculated based on monomers. The analysis of the acquired data was performed with the DiscoverMP software by applying a pre-measured standard (Refeyn Ltd., Oxford, UK). Counts were visualized in mass histograms as relative counts, which were calculated for the Gaussian fits of the measured peaks. For the determination of K_D-values and creating sigmoidal fitted curves, RStudio (RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA URL) was used.

Cryo-electron sample preparation and Data collection

Purified GS at a concentration of 1.5 mg/mL was rapidly applied to glow-discharged Quantifoil grids, blotted with force 4 for 3.5 s, and vitrified by directly plunging in liquid ethane (cooled by liquid nitrogen) using Vitrobot Mark IV (Thermo Fisher Scientific, Waltham, Massachusetts) at 100% humidity and 4 °C. To overcome preferred orientation bias, 0.7 mM CHAPSO was added to prevent water-air interface interactions, consequently the concentration of the protein was increased to 6mg/ml. We added purified commercially synthesized sP26 (Davids Biotechnologie, Regensburg, Germany) to all samples, but the peptide did not stably bind under the observed conditions. Data was acquired with EPU in EER-format on an FEI Titan Krios G4 (Cryo-EM Platform, Helmholtz Munich) equipped with a Falcon IVi detector (Thermo Fisher Scientific, Waltham, Massachusetts) with a total electron dose of ~55 electrons per Å² and a pixel size of 0.76 Å. Micrographs were recorded in a defocus range of -0.25 to -2.0 µm. For details see **suppl. Table S2** [↗](#).

Cryo EM - Image processing, classification and refinement

All data was processed using Cryosparc ([Punjani et al., 2017](#) [↗](#)). Micrographs were processed on the fly (motion correction, CTF estimation). Using blob picker, 878,308 particles were picked, 2D-classified and used for ab initio reconstruction. Iterative rounds of ab initio and heterogenous refinement were used to clean the particle stacks. The final refinements yielded models with an estimated resolution of 2.39 Å sets at the 0.143 cutoff (suppl. Fig. S3).

An initial model was generated from the protein sequences using alphaFold ([Jumper et al., 2021](#) [↗](#)), and thereupon fitted as rigid bodies into the density using UCSF Chimera ([Pettersen et al., 2021](#) [↗](#)). The model was manually rebuilt using Coot ([Emsley et al., 2010](#) [↗](#)). The final model was subjected to real-space refinements in PHENIX ([Liebschner et al., 2019](#) [↗](#)). Illustrations of the models were prepared using UCSF ChimeraX ([Pettersen et al., 2021](#) [↗](#)). The structure is accessible under PDB: 8s59. For details see **suppl. Table S2** [↗](#).

Acknowledgements

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Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. We thank Sandra Schuller for useful discussions and help in preparing manuscript figures. GKAH was supported by the Max Planck Society.

Figure supplements

Figure S1

Affinity-purified Strep-GlnA₁ and size-exclusion-chromatography (SEC) of Strep-GlnA₁ after purification.

A: 1.5 μg (lane 1) and 3 μg (lane 2) Strep-GlnA₁ on a coomassie-stained 12 % SDS-Gel. **B:** Elution profile of Strep-GlnA₁ (black) and size standard (dashed line, molecular weights in italics). Size exclusion chromatography was performed on a Superose™ 6 Increase 10/300 GL column (Cytiva, Marlborough, USA) with a flow rate of 0.5 ml/min.

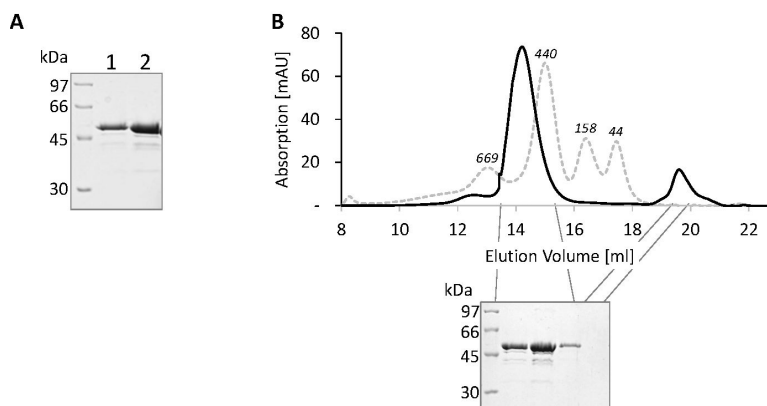
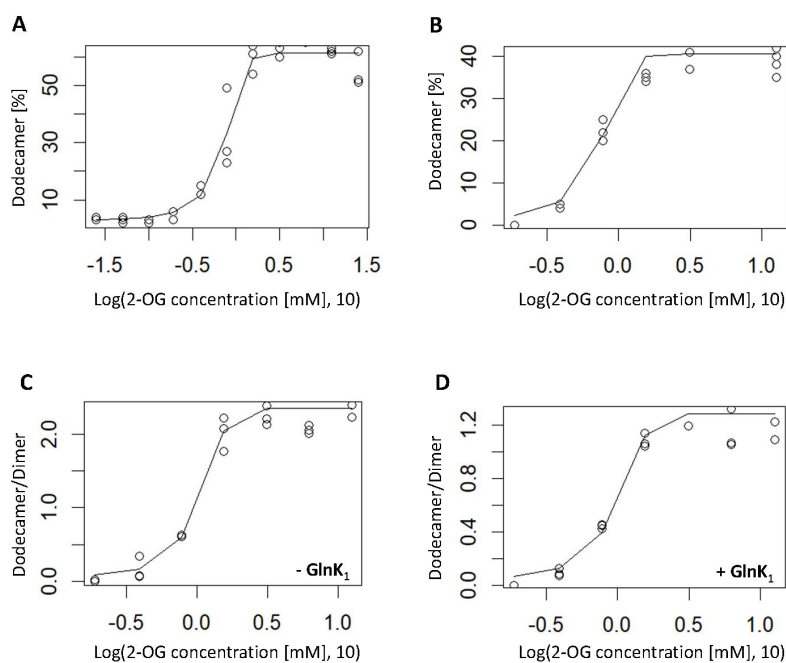


Figure S2

Sigmoidal fitted curves for mass photometry measurements of Strep-GlnA₁ with varying concentrations of 2-OG.

The curves were fitted and K_D -values calculated using RStudio (RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA URL). **A, B:** Two replicates for 2-OG titration, formation of dodecamer is shown in percent. **C, D:** 2-OG titration in the absence (C) and presence of GlnK₁ (D), formation of dodecamer is shown as a ratio of dodecamer/dimer.



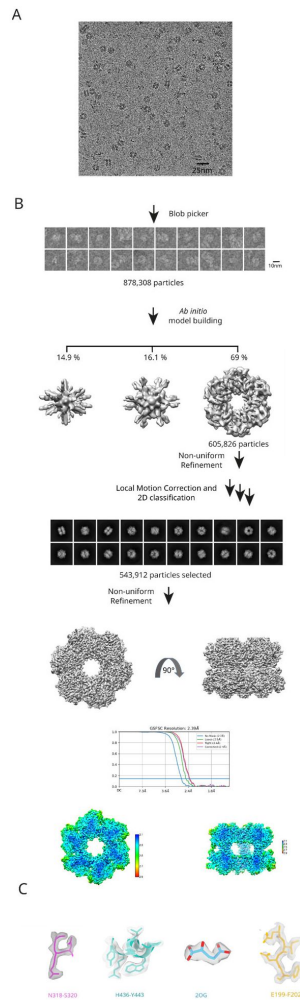


Figure S3

Cryo-EM Data processing workflow.

A: Representative motion-corrected micrograph showing different orientations of the GlnA₁ particles **B:** Cryo-EM processing tree used for obtaining the high-resolution structure of GlnA₁. The map obtained is coloured by resolution, where the global resolution was estimated using GSFSC **C:** Different regions of GlnA₁ encased around the cryo-EM density.

Figure S4

Mass photometry of purified and thawed Strep-GlnA₁ before and after SEC.

Mass spectra of Strep-GlnA₁ samples with 0 and 12.5 mM after affinity-purification (blue ●, 0 mM and green ●, 12.5 mM 2-OG) and after SEC (0 mM 2-OG, grey ●).

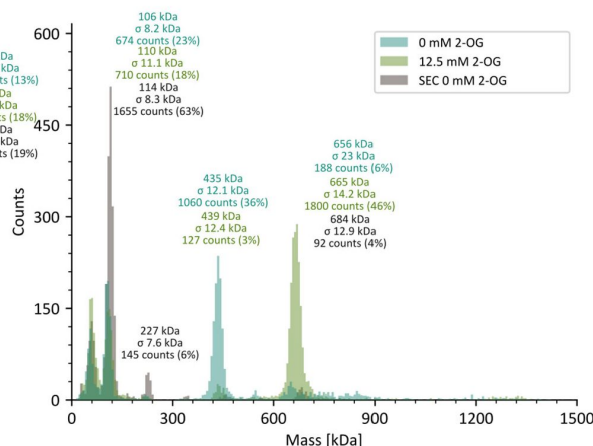
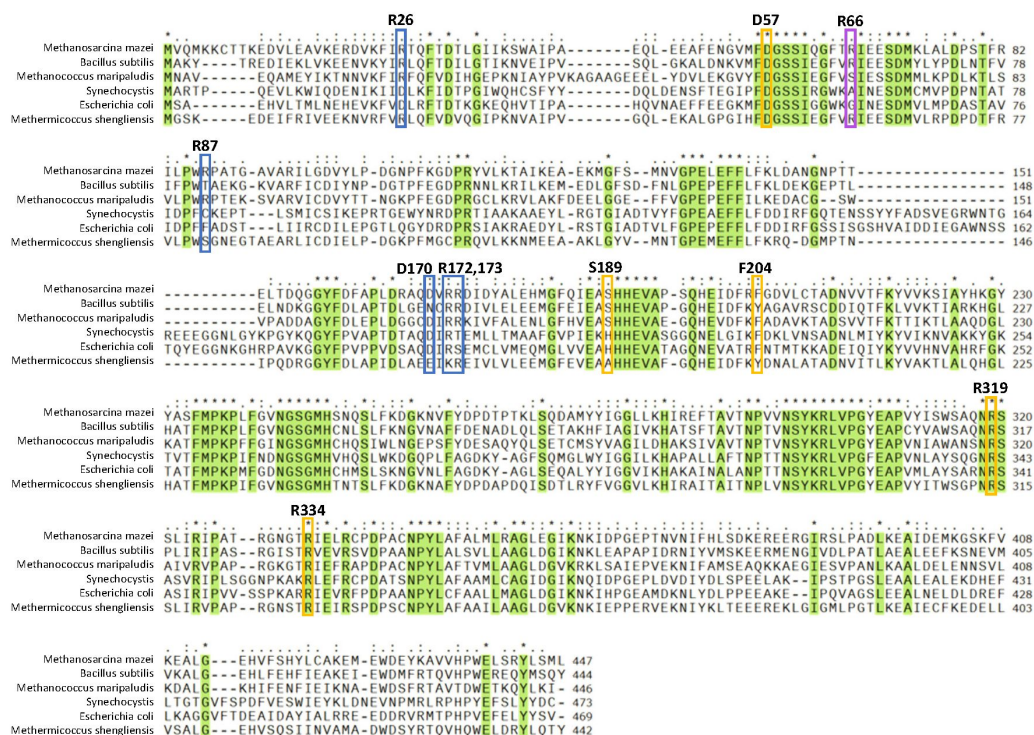


Figure S5

Amino-acid sequence alignment of different model organism glutamine synthetases.

(Alignment tool: COBALT, visualization in SnapGene) Conserved amino-acids are highlighted in green. The relevant residues in *M. maei* for 2-OG- and substrate-binding, as well as the arginine responsible for the feedback inhibition by glutamine are highlighted by coloured boxes (blue ●, orange ● and purple ●, respectively).



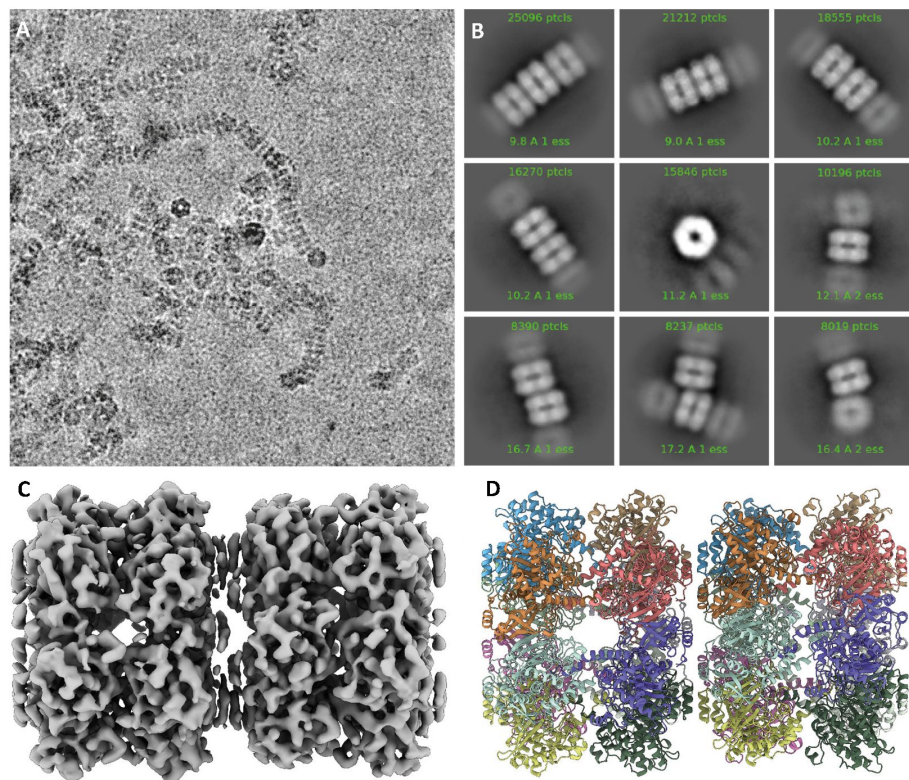


Figure S6

***M. mazei* GlnA₁ filaments.**

A: Representative motion-corrected micrograph showing GlnA₁ filaments **B:** Reference-free 2D classes showcasing filament orientations **C, D:** 3D reconstructed map of GlnA₁ filament and its model.

Supplementary tables

	Properties	Reference
Strains		
<i>E. coli</i> DH5 α	General cloning strain	(Hanahan, 1983)
<i>E. coli</i> BL21 (DE3)	Strain for protein expression	Thermo Fisher Scientific, Waltham, USA
<i>E. coli</i> BL21 (DE3) + pRIL	Strain for protein expression of genes with unusual codons/ Cm ^R	Stratagene, La Jolla, USA
Plasmids		
pET21a	General cloning vector providing a C-terminal His ₆ -tag	Novagen/Merck, Darmstadt, Germany
pETSUMO	Expression vector providing an N-terminal His ₆ -SUMO-tag	Thermo Fisher Scientific, Waltham, USA
pRS375	pET28a/Strep + GlnA ₁	(Gutt et al., 2021)
pRS1672	pETSUMO + GlnK ₁	This work
pRS1728	pEX-A258 + codon-optimized GlnA ₁ and sP26	Eurofins Scientific, Ebersberg, Germany
pRS1841	pRS375 + codon-optimized GlnA ₁	This work
pRS1863	pET21a + codon-optimized sP26	This work
pRS1951	pRS1840mut: R66AGlnA ₁	This work

Table S1

Strains and plasmids.

Dodecamer Gln complex	
(EMDB- 19730)	
(PDB 8S59)	
Data collection and processing	
Magnification	165,000 x
Voltage (kV)	300
Electron exposure (e-/Å ²)	55.0
Defocus range (µm)	0.25-2.0
Pixel size (Å)	0.72
Symmetry imposed	D6
Initial particle images (no.)	1,243,001
Final particle images (no.)	878,308
Map resolution (Å)	2.39
FSC threshold	0.143
Map resolution range (Å)	2.3-3.0
Refinement	
Initial model used (PDB code)	<i>de novo</i> , AlphaFold
Model resolution (Å)	2.5
FSC threshold	0.5
Model resolution range (Å)	2.2-2.5
Map sharpening <i>B</i> factor (Å ²)	-80.1
Model composition	
Non-hydrogen atoms	46968
Protein residues	5352
Ligands	AKG: 12
<i>B</i> factors (Å ²)	
Protein	16.58
Ligand	15.64
R.m.s. deviations	
Bond lengths (Å)	0.003
Bond angles (°)	0.602
Validation	
MolProbity score	1.29
Clashscore	5.32
Poor rotamers (%)	0.46
Ramachandran plot	
Favored (%)	98.22
Allowed (%)	1.56
Disallowed (%)	0.23

Table S2

Cryo-EM data collection, refinement and validation statistics

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Editors

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Reviewer #1 (Public Review):

Summary:

This study shows a new mechanism of GS regulation in the archaean *Methanosarcina mazei* and clarifies the direct activation of GS activity by 2-oxoglutarate, thus featuring another way in which 2-oxoglutarate acts as a central status reporter of C/N sensing.

Mass photometry and single particle cryoEM structure analysis convincingly show the direct regulation of GS activity by 2-OG promoted formation of the dodecameric structure of GS. The previously recognized small proteins GlnK1 and Sp26 seem to play a subordinate role in GS regulation, which is in good agreement with previous data. Although these data are quite clear now, there remains one major open question: how does 2-OG further increase GS

activity once the full dodecameric state is achieved (at 5 mM)? This point needs to be reconsidered.

Strengths:

Mass photometry reveals a dynamic mode of the effect of 2-OG on the oligomerization state of GS. Single particle Cryo-EM reveals the mechanism of 2-OG mediated dodecamer formation.

Weaknesses:

It is not entirely clear, how very high 2-OG concentrations activate GS beyond dodecamer formation.

The data presented in this work are in stark contrast to the previously reported structure of *M. mazei* GS by the Schumacher lab. This is very confusing for the scientific community and requires clarification. The discussion should consider possible reasons for the contradictory results.

Importantly, it is puzzling how Schumacher could achieve an apo-structure of dodecameric GS? If 2-OG is necessary for dodecameric formation, this should be discussed. If GlnK1 doesn't form a complex with the dodecameric GS, how could such a complex be resolved there?

In addition, the text is in principle clear but could be improved by professional editing. Most obviously there is insufficient comma placement.

<https://doi.org/10.7554/eLife.97484.1.sa2>

Reviewer #2 (Public Review):

Summary:

Herdering et al. introduced research on an archaeal glutamine synthetase (GS) from *Methanosarcina mazei*, which exhibits sensitivity to the environmental presence of 2-oxoglutarate (2-OG). While previous studies have indicated 2-OG's ability to enhance GS activity, the precise underlying mechanism remains unclear. Initially, the authors utilized biophysical characterization, primarily employing a nanomolar-scale detection method called mass photometry, to explore the molecular assembly of *Methanosarcina mazei* GS (*M. mazei* GS) in the absence or presence of 2-OG. Similar to other GS enzymes, the target *M. mazei* GS forms a stable dodecamer, with two hexameric rings stacked in tail-to-tail interactions. Despite approximately 40% of *M. mazei* GS existing as monomeric or dimeric entities in the detectable solution, the majority spontaneously assemble into a dodecameric state. Upon mixing 2-OG with *M. mazei* GS, the population of the dodecameric form increases proportionally with the concentration of 2-OG, indicating that 2-OG either promotes or stabilizes the assembly process. The cryo-electron microscopy (cryo-EM) structure reveals that 2-OG is positioned near the interface of two hexameric rings. At a resolution of 2.39 Å, the cryo-EM map vividly illustrates 2-OG forming hydrogen bonds with two individual GS subunits as well as with solvent water molecules. Moreover, local side-chain reorientation and conformational changes of loops in response to 2-OG further delineate the 2-OG-stabilized assembly of *M. mazei* GS.

Strengths & Weaknesses:

The investigation studies the impact of 2-oxoglutarate (2-OG) on the assembly of *Methanosarcina mazei* glutamine synthetase (*M. mazei* GS). Utilizing cutting-edge mass photometry, the authors scrutinized the population dynamics of GS assembly in response to varying concentrations of 2-OG. Notably, the findings demonstrate a promising and straightforward correlation, revealing that dodecamer formation can be stimulated by 2-OG

concentrations of up to 10 mM, although GS assembly never reaches 100% dodecamerization in this study. Furthermore, catalytic activities showed a remarkable enhancement, escalating from 0.0 U/mg to 7.8 U/mg with increasing concentrations of 2-OG, peaking at 12.5 mM. However, an intriguing gap arises between the incomplete dodecameric formation observed at 10 mM 2-OG, as revealed by mass photometry, and the continued increase in activity from 5 mM to 10 mM 2-OG for *M. mazei* GS. This prompts questions regarding the inability of *M. mazei* GS to achieve complete dodecamer formation and the underlying factors that further enhance GS activity within this concentration range of 2-OG.

Moreover, the cryo-electron microscopy (cryo-EM) analysis provides additional support for the biophysical and biochemical characterization, elucidating the precise localization of 2-OG at the interface of two GS subunits within two hexameric rings. The observed correlation between GS assembly facilitated by 2-OG and its catalytic activity is substantiated by structural reorientations at the GS-GS interface, confirming the previously reported phenomenon of "funnel activation" in GS. However, the authors did not present the cryo-EM structure of *M. mazei* GS in complex with ATP and glutamate in the presence of 2-OG, which could have shed light on the differences in glutamine biosynthesis between previously reported GS enzymes and the 2-OG-bound *M. mazei* GS.

Furthermore, besides revealing the cryo-EM structure of 2-OG-bound GS, the study also observed the filamentous form of GS, suggesting that filament formation may be a universal stacking mechanism across archaeal and bacterial species. However, efforts to enhance resolution to investigate whether the stacked polymer is induced by 2-OG or other factors such as ions or metabolites were not undertaken by the authors, leaving room for further exploration into the mechanisms underlying filament formation in GS.

<https://doi.org/10.7554/eLife.97484.1.sa1>

Reviewer #3 (Public Review):

Summary:

The current manuscript investigates the effect of 2-oxoglutarate and the Glk1 protein as modulators of the enzymatic reactivity of glutamine synthetase. To do this, the authors rely on mass photometry, specific activity measurements, and single-particle cryo-EM data.

From the results obtained, the authors convey that glutamine synthetase from *Methanosarcina mazei* exists in a non-active monomeric/dimeric form under low concentrations of 2-oxoglutarate, and its oligomerization into a dodecameric complex is triggered by higher concentration of 2-oxoglutarate, also resulting in the enhancement of the enzyme activity.

Strengths:

Glutamine synthetase is a crucial enzyme in all domains of life. The dodecameric fold of GS is recurrent amongst prokaryotic and archaea organisms, while the enzyme activity can be regulated in distinct ways. This is a very interesting work combining protein biochemistry with structural biology.

The role of 2-OG is here highlighted as a crucial effector for enzyme oligomerization and full reactivity.

Weaknesses:

Various opportunities to enhance the current state-of-the-art were missed. In particular, omissions of the ligand-bound state of GnK1 leave unexplained the lack of its interaction with

GS (in contradiction with previous results from the authors). A finer dissection of the effect and role of 2-oxoglutarate are missing and important questions remain unanswered (e.g. are dimers relevant during early stages of the interaction or why previous GS dodecameric structures do not show 2-oxoglutarate).

<https://doi.org/10.7554/eLife.97484.1.sa0>

Author response:

Reviewer #1 (Public Review):

We thank Reviewer #1 for the professional evaluation and raising important points. We will address those comments in the updated manuscript and especially improve the discussion in respect to the two points of concern.

(1) *How can GlnA1 activity further be stimulated with further increasing 2-OG after the dodecamer is already fully assembled at 5 mM 2-OG.*

We assume a two-step requirement for 2-OG, the dodecameric assembly and the priming of the active sites. The assembly step is based on cooperative effects of 2-OG and does not require the presence of 2-OG in all 2-OG-binding pockets: 2-OG-binding to one binding pocket also causes a domino effect of conformational changes in the adjacent 2-OG-unbound subunit, as also described for *Methanothermococcus thermolithotrophicus* GS in Müller et al. 2023. Due to the introduction of these conformational changes, the dodecameric form becomes more favourable even without all 2-OG binding sites being occupied. With higher 2-OG concentrations present (> 5mM), the activity increased further until finally all 2-OG-binding pockets were occupied, resulting in the priming of all active sites (all subunits) and thereby reaching the maximal activity.

(2) *The contradictory results with previously published data on the structure of M. mazei by Schumacher et al. 2023.*

We certainly agree that it is confusing that Schumacher et al. 2023 obtained a dodecameric structure without the addition of 2-OG, which we claim to be essential for the dodecameric form. 2-OG is a cellular metabolite that is naturally present in *E. coli*, the heterologous expression host both groups used. Since our main question focused on analysing the 2-OG effect on GS, we have performed thorough dialysis of the purified protein to remove all 2-OG before performing MP experiments. In the absence of 2-OG we never observed significant enzyme activity and always detected a fast disassembly after incubation on ice. We thus assume that a dodecamer without 2-OG in Schumacher et al. 2023 is an inactive oligomer of a once 2-OG-bound form, stabilized e.g. by the presence of 5 mM MgCl₂.

The GlnA1-GlnK1-structure (crystallography) by Schumacher et al. 2023 is in stark contrast to our findings that GlnK1 and GlnA1 do not interact as shown by mass photometry with purified proteins. A possible reason for this discrepancy might be that at the high protein concentrations used in the crystallization assay, complexes are formed based on hydrophobic or ionic protein interactions, which would not form under physiological concentrations.

Reviewer #2 (Public Review):

We thank Reviewer #2 for the detailed assessment and valuable input. We will address those comments in the updated manuscript and clarify the message.

| (1) *The discrepancy of the dodecamer formation (max. at 5 mM 2-OG) and the enzyme activity (max. at 12.5 mM 2-OG).*

We assume that there are two effects caused by 2-OG: 1. cooperativity of binding (less 2-OG needed to facilitate dodecamer formation) and 2. priming of each active site. See also Reviewer #1 R.1). We assume this is the reason why the activity of dodecameric GlnA1 can be further enhanced by increased 2-OG concentration until all catalytic sites are primed.

| (2) *The lack of the structure of a 2-OG and ATP-bound GlnA1.*

Although we strongly agree that this would be a highly interesting structure, it seems out of the scope of a typical revision to request new cryo-EM structures. We evaluate the findings of our present study concerning the 2-OG effects as important insights into the strongly discussed field of glutamine synthetase regulation, even without the requested additional structures.

| (3) *The observed GlnA1-filaments are an interesting finding.*

We certainly agree with the referee on that point, that the stacked polymers are potentially induced by 2-OG or ions. However, it is out of the main focus of this manuscript to further explore those filaments. Nevertheless, this observation could serve as an interesting starting point for future experiments.

| **Reviewer #3 (Public Review):**

We thank Reviewer #3 for the expert evaluation and inspiring criticism.

| (1) *Encouragement to examine ligand-bound states of GlnK1.*

We agree and plan to perform the suggested experiments exploring the conditions under which GlnA1 and GlnK1 might interact. We will perform the MP experiments in the presence of ATP. In GlnA1 activity test assays when evaluating the presence/effects of GlnK1 on GlnA1 activity, however, ATP was always present in high concentrations and still we did not observe a significant effect of GlnK1 on the GlnA1 activity.

| (2) *The exact role of 2-OG could have been dissected much better.*

We agree on that point and will improve the clarity of the manuscript. See also Reviewer #1 R.1.

| (3) *The lack of studies on dimers.*

This is actually an interesting point, which we did not consider during writing the manuscript. Now, re-analysing all our MP data in this respect, GlnA1 is likely a dimer as smallest species. Consequently, we will add more supplementary data which supports this observation and change the text accordingly.

| (4) *Previous studies und structures did not show the 2-OG.*

We assume that for other structures, no additional 2-OG was added, and the groups did not specifically analyse for this metabolite either. All methanoarchaea perform methanogenesis and contain the oxidative part of the TCA cycle exclusively for the generation of glutamate (anabolism) but not a closed TCA cycle enabling them to use internal 2-OG concentration as

internal signal for nitrogen availability. In the case of bacterial GS from organisms with a closed TCA cycle used for energy metabolism (oxidation of acetyl CoA) like e.g. *E. coli*, the formation of an active dodecameric GS form underlies another mechanism independent of 2-OG. In case of the recent *M. mazei* GS structures published by Schumacher et al. 2023, the dodecameric structure is probably a result from the heterologous expression and purification from *E. coli*. (See also Reviewer #1 R.2). One example of methanoarchaeal glutamine synthetases that do in fact contain the 2-OG in the structure, is Müller et al. 2023.

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